Boston Studies in the Philosophy and History of Science 325

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Encouraging Openness

Essays for Joseph Agassi on the Occasion of His 90th Birthday

Springer

Chapter 12
Francis Bacon, Karl Popper, Michael Polanyi, and Modern Experimental Biology

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Francis Bacon, widely appreciated as the father of the experimental method in science, proposed that science is inductive. Popper put critical testing of hypotheses against empirical evidence at the center of his epistemology, thereby excluding the act of conceiving or inventing a hypothesis or theory from his logical analysis of science. Michael Polanyi considered science as a social system based on epistemic authority and apprenticeship, stressing the informal and personal aspects of science. Using cases from genetics, embryology, molecular biology, and genomics, this essay (i) shows that induction, scientific authority and critical theory testing have been integral parts of scientific practice in the history of modern biology since the nineteenth century. This is despite the fact that a rigid adherence to Popper's principle of falsifiability is irreconcilable with scientific practice. (ii) analyzes the change of epistemology within recent developments in biology, particularly big data genomics. In these induction plays a major role while Polanyi's notions of epistemic authority and apprenticeship as well as Popper's principles of hypothesis, prediction and experimental testing are largely marginalized. The consequences of their disregard are discussed.

12.1 The Current Trend of Purely Inductive Approaches

Herbert Simon, computer scientist, cognitive philosopher, creator of the first artificial intelligence program and a recipient of the 1978 Nobel Prize in Economics in the 1970s successfully created a program that rediscovered Kepler's third law of

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N. Bar-Am, S. Gattei (eds.), *Encouraging Openness*, Boston Studies in the Philosophy and History of Science 325, DOI 10.1007/978-3-319-57669-5_12

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planetary motion: He explained: "We took raw data, nothing else, and tried to find patterns. We called it BACON, in honor of Sir Francis, because it's inductive. [...] In three tries BACON got the answer." (Cited in Stewart 1994)

Induction seems to have become fashionable again not only in artificial intelligence research, which is not the topic of this article, but also in new developments in the life sciences. With the advent of the generation of large amounts of data and high-throughput technology, the death of so-called hypothesis-driven science has recently been announced by scientists and commentators alike, for example journalist and editor of the Magazine "Wired" Chris Anderson. In a widely-cited article entitled "The end of theory", he proclaimed that big data should govern the science of the future, following the philosophy of Google: "Google conquered the advertising world with nothing more than applied mathematics. It didn't pretend to know anything about the culture and conventions of advertising – it just assumed that better data, with better analytical tools, would win the day. And Google was right." Anderson concluded that the received scientific method based on generating hypotheses and testing them experimentally had become obsolete, as had the acquisition of any knowledge prior to the application of an algorithm. Likewise, scientists' distinctions between correlation and causation and their aim to find causal mechanisms for observed phenomena should be a matter of the past, replaced by data-driven correlations (Anderson 2008). This implies not only the abandonment of a research which is at least in part Popperian, but also the premises that Polanyi deemed essential for successful research, i.e. competence, epistemic authority and apprenticeship.

Discovery science, "a scientific methodology which emphasizes analysis of large volumes of experimental data with the goal of finding new patterns or correlations, leading to hypothesis formation and other scientific methodologies" fits this trend in which it is data that drive science and lead to discoveries (Discovery science 2016). With data mining as the most common tool, discovery science is applied to data from diverse fields of study such as DNA analysis, climate modeling, nuclear reaction modeling, and others. As in Anderson's vision, prior knowledge is no longer required; data mining proceeds with the help of specialized machine learning algorithms for automated hypothesis forming and automated theorem proving. There seems to be no room for a creative and critical scientist.

The creation of a new belief system, such as the belief in algorithms without subsequent critical experimental testing of the hypotheses that were generated by the computer, deviates fundamentally from conventional research and the philosophical premises that guided it. The idea that observational data themselves can lead to discovery may be reminiscent of the notion of science as an inductive enterprise that was established by British philosopher and politician Francis Bacon (1561–1626), in particular of Bacon's thesis of the possibility of unbiased observations, his contempt for speculations or hypotheses, and his advocacy for new observations independent of any theory. (For a critical appreciation of Bacon and his philosophy of science, see Agassi 2013). However, the idea of observations leading directly to discovery disregards the fact that for Bacon, too, data are not to be obtained arbitrarily, 'groping, as of men in the dark', but guided by reason:

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> But the true method of experience, on the contrary, first lights the candle, and then by means of the candle shows the way; commencing as it does with experience duly ordered and digested, not bungling or erratic, and from it educing axioms, and from established axioms again new experiments. (Coveney et al. 2016)

The activities advocated by Anderson and Discovery science have been denounced as not meeting the criteria of science (Pigliucci 2009). However, independently of whether or not we call them science, research along these lines is increasing at public and private scientific institutions. That is, many future scientists will be united by a belief in big data, carry out correlations in genomic medicine, forget about hypotheses, prediction, experimental perturbations, causal-mechanistic explanations, and basic science in general. History shows that the kind of science which led to our current understanding of the natural world, sometimes called traditional science, flourished only in certain political environments, while either not existing at all or declining in others. A well-known example for the latter is the destruction of genetics in the Soviet Union for decades under Stalin's protégé Lysenko, who imposed his own holistic, non-causal-mechanistic principles on all research of heredity that, moreover, had to be oriented towards agricultural or medical applications.

This article is not critical of the use of big data, which not only strongly impacted upon medical and other applied research but also opened up new possibilities for basic research. Rather, the issue at hand is the proclaimed end of so-called conventional science. Most researchers combine big data research with conventional research guided by the testing of hypotheses. An example is Hans Lehrach, a pioneer in genomics, who made it clear (relating to Popper) that "we are not really talking about data versus hypotheses, since in all cases I can imagine we will need both, data and hypotheses. You can therefore either start with a hypothesis, and then generate the data to disprove it (that's all you can ever do with hypotheses anyway) or you first generate (maybe at higher cost) the data allowing the cost-effective testing of many hypotheses and then test many hypotheses on the available data (often gaining the enormous advantage that your power to disprove hypotheses is much higher). [...] In most real projects we keep alternating between data generating hypotheses generating the need to generate new data generating new hypotheses, etc. in a multistage process." Lehrach is critical of big-data projects that are not combined with "scientific evaluation that is based on more than genomic information" (Lehrach 2016).

Generally, most big-data projects are related to medical research. Researchers focus on mathematical tools and do not proceed to prediction and experimental testing of the computer-generated hypotheses. This is especially true of large enterprises such as the Broad Institute of MIT and Google Genomics that store and process enormous datasets and use tools to analyze such data in order to solve problems related to human health. These tools cannot be discussed, but critical hypothesis testing indeed seems to be a matter of the past.

In the remaining parts of this article I will use case studies from the history of what may be called "causal-mechanistic experimental biology" to examine researchers' underlying epistemology. These studies deal with fundamental questions related

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to development and heredity from the nineteenth until the twenty-first century. The focus is on analyzing the roles of inductive and hypothetico-deductive reasoning, as well as on the roles of competence and scientific belief.

12.2 Mechanistic Biologists from the Nineteenth to the Twenty-First Century and Their Epistemology

Experimental biology based on rational as well as empirical principles developed in the late nineteenth century. The uncritical belief in philosophical concepts such as those of the romantic school of natural philosophers, prevalent for example in idealistic morphology and early versions of cell theory (Canguilhem [1952] 2009, 105f.), became increasingly replaced by a search for causal mechanisms based on observation, critical reasoning and experimental testing.

Most of the discoveries that had a lasting impact on the development of genetics and molecular biology started with observations and inductive reasoning. As is well-known, induction as scientific method, as established by Bacon, related to reasoning that derived general theories or laws from specific empirical observations of natural events. Bacon argued that this could be achieved when a methodical and skeptical approach was used. His non-metaphysical approach rejected speculation and marginalized hypotheses, though it involved the guidance of reason, and was revived and developed further in the nineteenth century by British philosopher, scientist, and historian of science William Whewell (1794–1866). Whewell sought a 'middle way' between pure rationalism, which solely relies on deductive scientific reasoning without empirical tests and widespread ultra-empiricism. According to Whewell, all knowledge has an ideal (subjective) as well as an objective (empirical) dimension. For him, induction was not just an enumeration of instances but contained an act of thought, "a New Element added to the combination [of instances] by the very act of thought by which they were combined" (Snyder 2012).

An example for such a reasoned generalization of observations, followed by the generation of hypotheses, predictions, and experimental hypothesis testing is the work of Gregor Mendel on plant hybrids which led him to propose the two laws of hybridization that became the basis for the science of genetics. He explained his procedure as follows (1866):

> Experience of artificial fertilization, such as is effected with ornamental plants in order to obtain new variations in color, has led to the experiments which will here be discussed. The striking regularity with which the same hybrid forms always reappeared whenever fertilization took place between the same species induced further experiments to be undertaken, the object of which was to follow up the developments of the hybrids in their progeny.

To him, quantification was essential: "Among all the numerous experiments made, not one has been carried out to such an extent and in such a way as to make it possible to determine the number of different forms under which the offspring of

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the hybrids appear, or to arrange these forms with certainty according to their separate generations, or definitely to ascertain their statistical relations."

Mendel did not start from an existing theory on the formation of hybrids and their statistical relationships. In contrast to most of the workers in the field of hybridization, he took a methodical and unbiased approach: He was not guided (as was, for example, Darwin) by the widespread expectation that inheritance would proceed by blending the characters of the parents and other popular myths. Instead, he used methods that had been developed in the hard sciences and statistics, including the start from clearly defined initial conditions. The hypothesis with which he started his experiments, namely that hybridization follows statistical laws, was based on many observations and the inductive reasoning that the rules he hoped to find in experiments with one differentiating pair of characters, could be extended to other differentiating pairs of characters and would turn out to be valid for other species as well. While he experimentally tested his predictions for *Pisum* and *Phaseolus*, he encouraged other researchers to test them as well in other species; he demanded that his own experiments be "repeated and verified" (Herskowitz 1962).

The novelty of Mendel's empirical approach, contrasted with that of many of his colleagues who were guided by philosophically motivated 'rational' theories, becomes clear in his correspondence with Carl Naegeli, botanist at the University of Munich. Naegeli criticized Mendel for his alleged lack of a theoretical basis: "You should regard the numerical expressions as being only empirical, because they cannot be proved rational." (Herskowitz 1962) In his response Mendel made it clear that his empirical approach was the basis for general statements and the formulation of quantitative laws.

According to Sander Gliboff (1999), Naegeli's use of the term "rational" was probably a reference to the rational induction method of his mentor Matthias Schleiden, a German botanist and co-founder of cell-theory, and other neo-Kantians, who concluded general laws and explanations from regularities in empirical observations with the aid of a priori assumptions (such as time, space, and causality). Empirical testing was not involved. In addition, Mendel's hypotheses of intracellular elements and mechanisms of combinations did not meet Schleiden's guiding maxims for botany that called for explanations in terms of cells and developmental processes at the cellular level and therefore, in this sense, were not rational (ibid.).

Naegeli might have disliked Mendel's approach not only because of its contradiction with Schleiden's a priori assumptions, but also for the fact that Mendel reduced complex phenomena to simple particulate models, thus rendering a quantitative analysis fruitful. This stood in stark contrast to Naegeli's predilection for concepts of gradation and continuity.

Mendel's methods became a model for scientists, as Thomas Hunt Morgan, the founder of the school of classical genetics explained (1934): "[Mendel's] analysis was a wonderful feat of reasoning. He verified his reasoning by the recognized experimental procedure of science." This procedure is characterized by observation – induction – hypothesis – prediction – experimental testing.

Cytologist and embryologist Theodor Boveri's work, which laid the cytological basis for the science of genetics and was, in addition, crucial for the concept of the

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causal role of chromosomes for development, is a pertinent example for Whewell's 'consilience of induction' in biology. Whewell introduced this term to discuss the unification of knowledge between the different branches of learning: Evidence from independent fields of research can converge to reach strong conclusions.

In 1888, based on observations that the arrangement and shape of chromosomes is preserved in the daughter cells, Boveri formulated his theory of chromosome continuity and individuality, namely that chromosomes are continuous entities, different from one another, that exist throughout the cell cycle, condensed during division and relaxed between divisions (Maderspacher 2008). This theory included a bold conjecture because chromosomes become invisible in the microscope between cell divisions. A few years later, after conducting perturbation experiments which suggested that only a full set of chromosomes could support development, he proposed (at the same time as Walter Sutton) the idea that chromosomes are carriers of heredity, which soon became known as the chromosome theory of inheritance. The theory was heavily debated at the time, and even rejected by some, among them Thomas H. Morgan, who became its strongest supporter after he confirmed it with his own work in Drosophila.

Boveri's underlying epistemology up to this point can be characterized, as with Mendel, as reasoned generalization of observations, followed by the generation of hypotheses, and their experimental testing. The next step was consilience: Boveri considered the chromosome theory confirmed – in a probabilistic way – when results obtained independently in two different fields of research agreed with each other: "We now realize that two areas of research [cytology and genetics] which developed independently from one another, produced results, which are fitting together so exactly as if they were derived theoretically from one another. [...] Thus the probability that the characters of Mendel's experiments are indeed linked to certain chromosomes is exceptionally great." (Boveri 1904, p. 123, cited after Barthelmess 1952, p. 233, trans. UD)

Around 50 years later, biochemist Erwin Chargaff proposed, purely inductively, a statistical rule in the composition of DNA namely that the ratio of Adenin to Thymin and that of Guanin to Cytosin in DNA nearly equals one. When he started to scrutinize the composition of the DNA of different species with newly developed chromatographical methods, he did not expect any regularities in the composition of DNA. On the contrary: He examined DNA because he was one of the few researchers at the time who believed Oswald T. Avery's demonstration that DNA, and not protein, was the hereditary material. Therefore he tested the hypotheses that DNA did not consist of regularly repeating sequences and that the sequences of DNAs from different species differed from one another.

While confirming these hypotheses, he unexpectedly found a pattern in the sequences, the rule in the ratios of bases, which would soon carry his name. After further measurements in various other species this was soon accepted as a general rule by inductive reasoning. The elucidation of the DNA double helix structure by Watson and Crick in 1953 made it clear that the rule had to hold universally, because the structure of DNA was accepted to be universally valid (Godfrey-Smith 2008).

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All scientists mentioned before, to my knowledge did not relate to any epistemology as guiding principle, with the exception Schleiden and Schwann relating to Kant. The above makes it clear that all of them regarded empiricism highly, and that, in addition, they put forward hypotheses which they tested experimentally. The hypothetical-deductive method, which placed hypothesis testing in the center of its epistemology, was formulated most forcefully by Austrian-British philosopher Karl Popper (though it was partly anticipated by Whewell). Popper's epistemology does not need to be outlined here; I briefly summarize only those aspects that are relevant for my subsequent analysis. According to him, a scientist puts forward statements and tests them step by step. In empirical science, he constructs hypotheses or systems of theories and critically tests them against experience by observation and experiment. The riskier a prediction is, the better is the test of the hypothesis. This testing is deductive, that is, it does not follow an inductive logic (Popper [1959] 2002).

Popper rejected induction as a way to test hypotheses because he perceived an asymmetry between confirmation and falsification. No finite set of observations, for example of white swans, can prove the statement that all swans are white: "It is usual to call an inference 'inductive' if it passes from singular statements [...], such as accounts of the results of observation or experiments, to universal statements, such as hypotheses or theories. Now it is far from obvious from a logical point of view, that we are justified in inferring universal statements from singular ones, no matter how numerous; for any conclusion drawn in this way may always turn out to be false: no matter how many instances of white swans we may have observed, this does not justify the conclusion that all swans are white." (Ibid., 4)

Popper also rejected induction as a necessary means for creating hypotheses. Dismissing the accumulation of more and more perceptual experiences as basis for the advance of science, he contradicted Bacon: "Bold ideas, unjustified anticipations, and speculative thought, are our only means for interpreting nature: our only organon, our only instrument for grasping her" (ibid., 280). He held that not only truth cannot be achieved in such a way but also that probability is impossible: "I never assume that by force of 'verified' conclusions, theories can be established as 'true', or even as merely 'probable'." In contrast, theories can be falsified by a single piece of contradicting evidence. Therefore falsifiability, that is, the inherent possibility that a hypothesis can be proven false, not induction and inference, is at the center of Popper's scientific methodology. Theories can only be refuted, not confirmed. A theory that has been falsified must be rejected. According to Nimrod Bar-Am (2014), Popper was aware of the fact that the majority of scientists did not engage in creating risky refutable hypotheses but rather followed more conventional ways, which Thomas Kuhn (1962) called 'normal science'.

I do not present here another attempt to interpret Popper, rather I try to point out the Popperian and non-Popperian nature of experimental biologists' practice. Such a reflection has a new relevance at a time in which, as outlined above, entirely non-Popperian biology has become fashionable.

In two major respects, Popper's philosophy of science is not compatible with the actual work of the scientists depicted here. The first is related to Popper's refusal to

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analyze how hypotheses are set up or discoveries made (despite the misleading title *logic of discovery* of his 1959 book). The second is falsification. Popper's interest was in evaluating scientific theories. In a section on 'Elimination of Psychologism', he outlined what, in his opinion, philosophy of science should focus on (1959, 31): "The work of the scientist consists in putting together and testing theories. The initial stage, the act of conceiving or inventing a theory, seems to me neither to call for logical analysis nor to be susceptible of it. The question how it happens that a new idea occurs to a man – whether it is a musical theme, a dramatic conflict, or a scientific theory – may be of great interest to empirical psychology; but it is irrelevant to the logical analysis of scientific knowledge."

According to immunologist and Nobel laureate Peter Medawar, "Popper's most high-profile philosophical disciple within the field of active science" (Calver 2013), "the weakness of the hypothetico-deductive system, in so far as it might profess to offer a complete account of the scientific process, lies in its disclaiming any power to explain how hypotheses come into being" (Medawar [1963] 1996, 32). Joseph Agassi (2013, 74–75) emphasized that, according to Popper, novelty is relative to the theory that it refutes and new observations are viewed as refutations of received theories. In the history of biology, this is, however, rarely the reason for novelty, though there are examples such as the refutation of proteins as genetic material and its replacement by DNA, which marked the beginning of a drastic change in concepts and methods of molecular genetics (Deichmann 2004).

Unlike Popper suggested, the successful early research in genetics and development began with an inductive process in the sense of Whewell, followed by hypothesis. Observation and the accumulation of experiments was crucial, and sometimes new hypotheses were made by observations without a theory, as in the case of Chargaff's rule. Consilience of induction as in the case of Boveri, was the basis of major novelties. In general, progress in biology occurs predominantly through the opening up of new fields and not through rejected theories. A prominent example is the rise of molecular genetics which did not reject any theories of the existing classical genetics; it occurred after the concept of macromolecules was introduced into chemistry. The availability of new technologies often gave rise to new hypotheses, as did the microscope and staining to cell theory or heavy and radioactive isotopes to molecular biology.

While for experimental scientists testability and therefore falsifiability was a self-evident characteristic of hypotheses, they tried in most cases, as expressed by Morgan, to verify them; they certainly did not aim at falsifying them and sometimes avoided experiments to that effect. Thus, even if according to Popper falsifying could be carried out unintentionally, this is certainly often not the case. As mentioned before, verification often did not proceed by simply generalizing repeating individual observations or experimental results, but by finding concurring results obtained by different methods. Another major difference with Popper's epistemology lies in the fact that hypotheses that were disproved usually were modified and not discarded, as in the case of Mendel, whose rules were later shown to apply only to those cases in which genes were located on different chromosomes and were not separated by crossing over. This led to the amendment of Mendel's rules, not their

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rejection. Knowing about the fallibility of all theories and models, scientists nevertheless aimed at finding the truth about certain mechanisms. The degree of confirmation was often assessed in probabilistic terms.

This discrepancy between the philosophy underlying researchers' practice – which included induction and probability and often excluded the discarding of hypotheses once they were falsified – and that of Popper should not, however, obscure the fact that prediction and critical testing of hypotheses were fundamental principles in the search for causes and mechanisms of basic life phenomena.

At the same time that biology laid the basis for an understanding of the mechanisms of development and its genomic control, research was conducted on the basis of entirely different concepts. Here I briefly present research in which, about a century before the advent of big data genomics at the end of the twentieth century (though without major impact), mathematical tools and correlations were predominant. An example is the mathematical-morphological approach to understanding life phenomena by the British zoologist Wentworth d'Arcy Thompson who was influenced by Platonic philosophy. In his widely read book *On Growth and Form*, first published in 1917, he applied simple mathematics in order to find unifying principles of life's forms. According to Thompson, organic form was predetermined by the physical organization of the system in which it developed, and obeyed the law of physics, in which form was a diagram of forces. Theories that attributed specific properties to particles of the protoplasm, such as chromosomes, were rejected. Thompson's aim was "to correlate with mathematical statement and physical law certain of the simpler outward phenomena of organic growth and structure or form" (Thompson [1917] 1942, p. 14). He regarded his task to be finished if he had been able "to show that a certain mathematical aspect of morphology, to which as yet the morphologist gives little heed, is interwoven with his problems, complementary to his descriptive task, and helpful, nay essential, to his proper study and comprehension of Growth and Form" (ibid., p.1096).

With mathematical tools, Thompson created forms that correlated with certain forms or patterns of organisms, but he did not examine whether his mathematics represented causal relations in reality. Without perturbations, predictions, and experimental hypothesis testing, he could not elucidate mechanisms, apart from the fact that he neglected basic principles of life, in particular biological specificity (see, for example, Loeb 1916). Thompson's ideas did not play a role in the lasting advances of biology such as genetics, experimental embryology, and biochemistry though his approach has been taken up by mathematicians (see e.g. Katz 2012).

Starting in the 1960s, molecular biological research, which had previously focused on micro-organisms and viruses, broadened its scope to include the search for molecular explanations of phenomena that only occur in higher organisms, e.g. development and certain diseases. New techniques enabled systems approaches, e.g. through the study of the expression of a large number of genes at the same time. As pointed out in the beginning, this led to research that, on the one hand, analyzes large volumes of experimental data with the aim of finding new patterns or correlations, and in which hypothesis formation becomes an automated, purely inductive process conducted by computers. On the other hand, systems approaches pursue the

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same mechanistic goals as twentieth century experimental biology. An example is the research by Eric Davidson and his collaborators at Caltech on gene regulatory networks in development, a concept created by Davidson. A world leader in molecular embryology, Davidson demonstrated that, at least in sea urchins, early development is entirely regulated by the genome.

In 2006 he introduced the term 'regulatory genome' for the interactions between regulatory genes and their products during development. In the preface of his book with the same title Davidson wrote: "I have not shied away from what are sometimes pejoratively been termed 'big ideas' nor taken the view of an anonymous reviewer of a paper of mine who recently amused me with the complaint, 'But the original ideas in this paper are speculative!' This book includes many diagrams in which concepts are set forth in specific form, just so that they can be subjected to precise tests of falsification, and just so they can be used in precise ways to generate predictions I may not have thought of." (Davidson 2006, ix–x)

With the proposal of bold hypotheses that can be subjected to falsification, Davidson appears to be an ideal-type Popperian scientist. However the idea of the 'regulatory genome' was conceived and developed through decades-long painstaking experimental research by Davidson and his collaborators and anything but a one-step bold hypothesis. In this work Davidson and his collaborators systematically examined the cell-type specific gene expression patterns before moving on from the "gene-by-gene characterization of the sea urchin embryo to full comprehensiveness." (Rothenberg 2016) This systems approach was made possible when sequencing data of the whole sea-urchin genome were available.

Observations were crucial as a starting point. According to Davidson (2016), descriptive observations, often derided as intrinsically inferior to "problem solving," received a new meaning in a systems approach: "The whole interaction system cannot be studied effectively until the whole set of its components and the output (behavior) is correctly measured/observed." Following the acquisition of observational data, hypotheses were indispensable: "Throughout my own career in bioscience, I have exploited the explanatory effectiveness of hypothesis-driven scientific logic to explore how animals attain their properties, i.e., express their genomes and develop." This perturbation analysis is essential because "only by deliberate experimental perturbation and predictive challenge of the system can the mechanisms by which it operates be revealed." (Ibid.)

In order to test the proposed network, Davidson and his collaborators transformed it into a computational engine that would generate predictions of when and where in the embryo every regulatory gene should be expressed or not expressed according to the network structure. They showed that computation and observation agree in most cases (Peter et al. 2012).

Davidson's work displays clear Popperian traits, in particular with his strong emphasis on hypothesis (even bold hypotheses), predictability, and hypothesis-testing. His later attempts to include all regulatory genes, their products and all interactions into his study also seem to bear a Popperian mark: Davidson's outlining of a mechanism that takes care of any possibility, or, as he put it, without leaving

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any room for the appearance of 'black swans', appears to be inspired by his wish to overcome Popper's prohibition regarding induction.

But as with the mechanistic biologists before him, Davidson's Popper is not the Popper of philosophical textbooks. His statement that "in basic science, hypothesis formation is an inferential, inductive, creatively novel proposition of mechanism" (Davidson 2016) shows most clearly his emphasis on the relevance of induction in creating hypotheses. Apart from this, he would not have agreed with Popper that all experimental tests of hypotheses were attempts at falsifying them, no matter what scientists consciously thought they were doing. He replaced Popper's falsifiability criterion for good science with predictability: "In science, conceptual predictability accruing from an analysis is the golden criterion of value, of progress in understanding. Predictability is the lynchpin of scientific knowledge." (2016)

In addition, Davidson, like most experimental biologists, believed in the correctness of those of his hypotheses that were supported by various experimental (and computational) methods. They would have strongly disagreed with Popper's conviction that by force of 'verified' conclusions, "theories can never be established as 'true', or even as merely 'probable'." As pointed out by Michael Polanyi (1963) and Thomas Kuhn (1962), this belief frequently prevented novelties from being implemented immediately. As emphasized by Polanyi (1963), it also prevented uncorroborated claims from being accepted too early.

12.3 Scientific Belief and Criticism, Popper, and Polanyi

Michael Polanyi (1891–1976), a physical chemist who became a philosopher of science, took a very different outlook on science than Popper. Polanyi considered science a social system based on competence, authority and apprenticeship. A comprehensive overview of Polanyi's life and work from a historical-sociological and philosophical perspective is in Nye (2011). Though he accepted that rules and principles were important when it came to assessments of discovery and verification, he made it clear that these rules would not enable anyone who was lacking the training and skills of a scientist to make any competent judgement: "It is here, in the course of discovery and verification, that the premises of science exercise their guidance over the judgment of scientists. It is obvious that no formulation of these premises ever proposed (or yet to be proposed) would have enabled a person lacking the special gifts and training of a scientist, competently to decide any of the serious uncertainties that have risen in the various controversial or doubtful issues [….]." (Polanyi [1958] 1962, 165.)

Polanyi's conclusion that scientists are not driven by doubt, but by a personal commitment to beliefs reflects his own experience as a scientist, and contributes to the understanding of scientific practice. For Polanyi, a great scientist is someone who is competent: "To acknowledge a person as a scientist – and even as a very great scientist – is merely to acknowledge him as competent in science, which admits the possibility that he was, or is, in many ways mistaken." (Ibid., 173)

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In Polanyi's view of scientific knowledge as a "stable system of belief", an interpretive "framework" or "paradigm" or "tradition" plays a major role" (Nye 2011, 262). This idea was also taken up by Thomas Kuhn, who popularized the term 'paradigm'. Unlike Kuhn and Popper, but in line with many scientists, Polanyi was a realist, convinced that science results in true knowledge despite the fact that it is based on social activities. The interpretative frameworks could be changed through major discoveries. What he called the "logical gap" between scientists working in different frameworks (anticipating Kuhn's concept of incommensurability (Jacobs 2002)), could sometimes be resolved in the course of scientific progress. Polanyi's example was the late-nineteenth-century controversy between Pasteur and the chemists about the nature of alcoholic fermentation. This was resolved by the new conception of intracellular enzymes, discovered by Eduard Buchner (Polanyi [1958] 1962, 150–157).

Polanyi's emphasis on scientific authority, competence, and apprenticeship has been frequently criticized, most fiercely by Joseph Agassi, who wrote (1981, 1–2):

> The prevalent philosophy of science, especially in the social sciences which should know better, is that of Michael Polanyi, especially in his vulgarized version offered by Thomas S. Kuhn, of scientists as a guild of masters and apprentices and a hierarchy of leaders who rule by conformity to fashion, which conformity makes thought-police quite redundant.

This criticism neglects that authority and apprenticeship were the backbone of every successful scientific school in the history of science. Polanyi's notion of scientists' commitment to belief is based on the notions of competence and the assessment of theories. His emphasis on the crucial role of scientific-authorities-as-epistemic-authorities includes the notion that these authorities can be shown to be mistaken, thus requiring a change in the scientific field, i.e. they include the notion of scientific criticism (see in particular Polanyi 1963). Therefore Polanyi's ideas of belief and authority differ fundamentally from the belief in the authority of big data, which neither require prior knowledge of the scientific field in question nor testing of the results against reality.

It should be added that Polanyi, similarly to Russell and Popper (Deichmann 2011), emphasized the contribution of scientific skepticism to freedom and social improvements in nineteenth century Europe (though he thought that an excess of doubt and skepticism later on became destructive): "What is more, scientific skepticism smoothly cooperated at first with the new passions for social betterment. By battling against established authority, skepticism cleared the way for political freedom and humanitarian reforms. Throughout the nineteenth century, scientific rationalism inspired social and moral changes that have improved almost every human relationship, both private and public, throughout Western civilization." (Polanyi 1966, 57)

Scientific skepticism also has a cultural role in the twenty-first century. For Davidson (2016), the mechanistic understanding of development and evolution was not only the only way to conduct science fruitfully but it also created a bulwark against irrational views such as those expressed by creationists. Scientific skepticism is crucial in regard to claims of big data genomic consortia that are not based

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on sufficient biological knowledge and experimental tests. An example is the ENCODE Consortium whose databases are widely appreciated by researchers but whose methods and interpretations as a scientific enterprise have been heavily disputed (see Morange 2014 and references).

Neither induction nor critical testing of hypotheses can explain scientific advance if taken alone; competence and the existence of a passionate commitment to scientific beliefs is another necessary prerequisite for understanding the development of science. To replace hypothesis, prediction and critical experimental theory testing with a belief in the self-explanatory function of algorithms, correlations and statistics means to give up the search for causes and mechanisms. This completely changes both the nature of science, and equally importantly, its cultural role as well.

Acknowledgement I thank Mary Jo Nye for her valuable comments on the manuscript.

References

Agassi, Joseph. 1981. *Science and society: Studies in the sociology of science*, Boston studies in the philosophy and history of science. Dordrecht/Heidelberg: Springer.

———. 2013. *The very idea of modern science. Francis Bacon and Robert Boyle*, Boston studies in the philosophy and history of science. Dordrecht/Heidelberg: Springer.

Anderson, Chris. 2008. *The end of theory: The data deluge makes the scientific method obsolete*. 23 June 2008. http://www.wired.com/2008/06/pb-theory/. Accessed 22 Nov 2016.

Bar-Am, Nimrod. 2014. The structure of scientific revolutions, fourth edition, 50th anniversary. *Philosophy of the Social Sciences* 44: 688. doi: 10.1177/0048393112473424 (http://pos.sagepub.com/content/44/5/688)

Barthelmess, Alfred. 1952. *Vererbungswissenschaft*. Freiburg: Alber.

Calver, N. 2013. Sir Peter Medawar: Science, creativity and the popularization of Karl Popper. *Notes and records*. The Royal Society for the History of Science. 22 May 2013. doi: 10.1098/rsnr.2013.0022.

Canguilhem, Georges. [1952] 2009. *Die Erkenntnis des Lebens* (trans. from French: La connaissance de la vie). Berlin: August Verlag.

Coveney, P.V., Edward R. Dougherty, and Roger R. Highfield. 2016. Big data need big theory too. *Philosophical Transactions of the Royal Society A*. doi: 10.1098/rsta.2016.0153.

D'Arcy Thompson, W. [1917] 1942. *On growth and form*, 2nd ed. Cambridge: Cambridge University Press.

Davidson, E.H. 2006. *The regulatory genome: Gene regulatory networks in development and evolution*. San Diego: Academic Press.

———. 2016. Genomics, "discovery science", systems biology, and causal explanation: What really works? *Perspect.Biol.Med.* 58: 165–181. doi:10.1353/pbm.2015.0025.

Deichmann, U. 2004. Early responses to Avery's et al.'s 1944 paper on DNA as hereditary material. *Historical Studies in the Physical and Biological Sciences.* 34 (2): 207–233.

———. 2011. Michael Polanyi on scientific authority and his criticisms of Popper and Russell. In *Yearbook of the Leo Baeck Institute*, vol. 56, 249–268. London: Oxford University Press.

Discovery science. 2016. https://en.wikipedia.org/wiki/Discovery_science. Accessed 25 Aug 2016.

Gliboff, S. 1999. Gregor Mendel and the laws of evolution. *History of Science* 37: 217–235.

uted@post.bgu.ac.il

Godfrey-Smith, P. 2008. Induction, Samples, and Kinds. For a collection of papers from the 2008 Inland Northwest Philosophy Conference. http://philosophy.fas.nyu.edu/docs/IO/11840/pgsinduction.pdf. Accessed 23 Nov 2016.

Jacobs, S. 2002. Polanyi's presagement of the incommensurability concept. *Studies in History and Philosophy of Science A*. 33: 105–120.

Herskowitz, I.H. 1962. *Genetics*. Boston: Little, Brown, Supplements.

Katz, E.A. 2012. Bridges between mathematics, natural sciences, architecture and art: case of fullerenes. Book of papers and extended abstracts of the 1st International Conference "Art, Science and Technology: Interaction between Three Cultures", ed. by T. Kravchuk, A. Groysman, C. Soddu, E. Colabella and G. Leisman, 60–71. Milano: Domus Argenia.

Kuhn, Thomas S. 1962. *The structure of scientific revolutions*. Chicago: University of Chicago Press.

Lehrach, H. 2016. Interview with Hans Lehrach by Ute Deichmann. Berlin, 9 February 2016.

Loeb, J. 1916. *The organism as a whole from a physicochemical viewpoint*. New York/London: Putnam's Sons.

Maderspacher, F. 2008. Theodor Boveri and the natural experiment. *Current Biology* 18 (R): 278–R286. doi:10.1016/j.cub.2008.02.061.

Medawar, P. [1963] 1996. Hypothesis and imagination. Times Lit. Suppl. (25 October 1963), in: Medawar, P. 1996. *The strange case of the spotted mice and other classic essays on science*. Oxford: Oxford University Press.

Mendel, G. 1866. Experiments in plant hybridization (Versuche űber Pflanzen-Hybriden). *Verhandlungen des naturforschenden Vereins in Brűnn*, *4*, 3–47, (Read at the meetings of February 8th, and March 8th, 1865; in English translation at mendelweb.org).

Morange, M. 2014. Genome as a multipurpose structure built by evolution. *Perspectives in Biology and Medicine 57*: 162–171.

Morgan, T.H. 1934. *Embryology and genetics*. New York: Columbia University Press.

Nye, Mary Jo. 2011. *Michael Polanyi and his generation: Origins of the social construction of science*. Chicago: University of Chicago Press.

Peter, I.S., E. Faure, and E.H. Davidson. 2012. Predictive computation of genomic logic processing functions in embryonic development. *Proceeding of the National Academy of Sciences USA* 109: 16434–16442. doi:10.1073/pnas.1207852109.

Pigliucci, M. 2009. The end of theory in science? *EMBO Reports* 10: 534. doi:10.1038/embor.2009.111.

Polanyi, M. [1958] 1962. *Personal knowledge – towards a post-critical philosophy*. Chicago: University of Chicago Press.

———. 1963. The potential theory of adsorption. *Science.* 141: 1010–1013. doi:10.1126/science.141.3585.1010.

———. 1966. *The tacit dimension*. Garden City: Doubleday.

Popper, K. [1959] 2002. *The logic of scientific discovery*. London/New York: Routledge.

Rothenberg, E.V. 2016. Eric Davidson: Steps to a gene regulatory network for development. *Developmental Biology* 412: S7–S19. doi:10.1016/j.ydbio.2016.01.020.

Stewart, Douglas.1994. *Herbert Simon on the mind in the machine*. http://www.omnimagazine.com/archives/interviews/simon/index.html. Accessed 23 Nov 2016.

Snyder, Laura J. 2012. "William Whewell". *The Stanford Encyclopedia of Philosophy.* Winter 2012 Edition. ed. Edward N. Zalta. http://plato.stanford.edu/archives/win2012/entries/whewell/. Accessed 23 Nov 2016.

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